

Multimodal Transformer-Based Gait Analysis and Deep Learning Model for Slip-Resistant Footwear Evaluation

Shaghayegh Chavoshian , *Member, IEEE*, and Atena Roshan Fekr , *Member, IEEE*

Abstract—Slips and falls remain a leading cause of injury, particularly in icy winter conditions and occupational settings with slippery surfaces. Several laboratory and field-based systems are designed to assess footwear slip resistance, including mechanical testers, tribometers, and human-centered gait-based assessments. While mechanical tests offer standardized slip resistance metrics for footwear, they often fail to reflect real-world variability in human gait, environmental conditions, and anthropometric diversity. On the other hand, although human-centered testing ensures ecological validity, it typically reports a single aggregated score for footwear derived from multiple participants, which obscures individual biomechanical differences and limits the ability to provide personalized assessments of slip resistance performance. This study introduces a transformer-based regression model that integrates: 1) temporal biomechanical features; 2) outsole images; 3) footwear metadata; and 4) anthropometric data to predict personalized slip resistance score of 84 footwear across 578 walking trials conducted by 18 human participants on two types of ice: wet and dry. The predictive performance of our multimodal deep learning model varied by surface condition, with substantially better results on dry surfaces (RMSE: $1.34^\circ \pm 0.34^\circ$, R^2 : 0.45 ± 0.09 , MAPE: $10.48\% \pm 3.58\%$) than on wet surfaces (RMSE: $2.34^\circ \pm 0.43^\circ$, R^2 : 0.47 ± 0.12 , MAPE: $21.45\% \pm 6.89\%$). The results demonstrate the potential of fusing biomechanical and outsole characteristics to reliably assess slip risk under varying environmental conditions. Additionally, they emphasize the need for adaptive and context-aware testing protocols and pave the way for the development of intelligent footwear assessment tools aimed at injury prevention.

Impact Statement—To prevent slips and falls, wearing slip-resistant footwear is one of the most effective solutions. Slip resistance is commonly evaluated using time-consuming and costly experimental testing systems. In this study, we introduced an AI-driven system that personalizes footwear slip-resistance assessment by integrating computer vision, biomechanical gait analysis, and footwear characteristics with up to 90% accuracy, while the model explains 45%–47% of variance in slip resistance outcomes. This technology enables manufacturers to design footwear tailored to diverse populations rather than applying one-size-fits-all ratings. The economic impact includes reducing injury costs and enabling smarter consumer choices through personalized safety recommendations. Socially, it addresses equity by accounting for age, body type, and movement differences that disproportionately affect vulnerable workers. Environmentally, improved testing reduces wasteful product cycles. This work transforms slip prevention from reactive accident investigation to proactive, personalized risk prediction, potentially preventing thousands of injuries annually.

Index Terms—Convolutional neural network (CNN), gait analysis, slip resistance, transformer model, winter footwear.

I. INTRODUCTION

SLIPS and falls are one of the most common causes of injuries, especially in winter conditions or workplaces with slippery surfaces [1], [2]. In occupational settings alone, slips account for 40%–85% of reported falls and injuries, with industries such as construction, healthcare, and food services facing disproportionately high risks [3], [4], [5], [6], [7]. The primary cause of slipping is inadequate traction between the footwear and the walking surface [8], [9], [10]. Therefore, footwear plays a critical role in slip prevention, making slip resistance testing an essential step in ensuring safety across different environments.

To assess footwear slip resistance, two main measurement systems are used: 1) mechanical tests; and 2) human-centered tests. Mechanical tests use machines to measure coefficients of friction (CoF) under simulated conditions. This involves mimicking heel strike or flat foot contact by applying a known vertical force to represent body weight, followed by sliding the footwear across a test surface at a consistent speed and angle [11], [12]. To reflect various real-world scenarios, the tests are often conducted at different inclination angles, commonly 7°

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or 11° , representing normal walking or more extreme movement conditions [13], [14], [15], [16], [17]. Recent advances in slip safety assessment have increasingly emphasized the integration of surface properties, environmental conditions, and data-driven modeling approaches. A previous study demonstrated that surface friction coefficients combined with tribological analysis could reliably predict slip safety risk [18], while another study showed that portable, human-centered friction measurements significantly outperform purely mechanical indicators [19]. Therefore, while mechanical tests provide standardized metrics such as CoF, they often fail to replicate real-world complexities, such as gait variability or adaptive balance strategies. Human-centered testing, on the other hand, incorporates natural biomechanical and cognitive factors, offering superior ecological validity [20], [21].

Albeit useful, human-centered testing faces significant variability due to biomechanical differences. Additionally, factors such as age-related balance decline and impaired postural control increase the risk of slipping [3], [22]. Studies have shown that human gait adjustments during slips, such as reduced stride length or altered heel-strike angles, are critical for predicting real-world slip outcomes [23], [24]. Studies have also examined biomechanical responses, specifically kinematic parameters such as velocities, accelerations, and joint moments before a slip event, as key indicators for predicting slip severity and underlying mechanisms of slipping [11], [12], [13], [14]. For example, previous literature found out that hazardous slips and fall events are often associated with increased step lengths [23], [25], [26]. In addition, previous research investigated the kinematics of heel strike during walking with a focus on its influence on slip resistance testing. Another study found that an increased foot angle might influence the contact area between the shoe and the floor, potentially affecting the frictional forces and increasing slip risk [26], [27]. This also aligns with previous studies that highlighted the importance of knee joint angles in understanding slip dynamics during walking [28]. Therefore, different participants with varying gait patterns may demonstrate varying performance outcomes when wearing the same footwear. Existing human-centered footwear testing methods, such as the maximum achievable angle (MAA) test [20], primarily rely on aggregated outcome measures (minimum score among participants), which may overlook the individual gait dynamics.

On the other hand, age, sex, and body mass index (BMI) are strongly associated with slip outcomes. Older adults often experience declines in neuromuscular function and slower reaction times, contributing to up to 40% higher slip risk [29], [30], [31], [32]. While some other studies reported no association or even higher slips and fall rates among younger workers [33], [34], [35], [36], a comprehensive understanding requires considering age alongside other key factors with diverse occupational roles. Sex may also influence slip outcomes due to physiological and biomechanical differences. Studies have shown that up to 50% of female workers may be at higher risk [33], [37]. Higher BMI is also associated with higher slip risks [38], which might be due to altered gait mechanics and increased ground reaction forces.

Therefore, this study aims to investigate the influence of these factors on human-centered test outcomes and then develop a novel measurement model with an embedded accurate predictive model to support more informed, safety-focused footwear recommendations.

II. METHODS

A. Human-Centered Testing Protocol

Our human-centered testing protocol was conducted in an environmental chamber that simulates controlled winter conditions with an icy floor, including adjustable ice and ambient air temperatures. The tests were carried out under two distinct ice conditions: wet ice characterized by air temperatures of 7°C – 10°C ; and ice surface temperatures between -0.1°C and 0.8°C , and dry ice characterized by air temperatures of 1.5°C – 2.5°C and ice surface temperatures between -4.3°C and -3.3°C [39], [40].

The testing platform can be tilted from 0° to 15° to provide uphill and downhill conditions. Participants begin walking on a level surface (0° inclination) while wearing the test footwear. The incline angle is progressively increased until the participant experiences their first slip event. Following the first slip, the inclination is decreased in 1° increments to determine the MAA, and that angle value is used as our quantitative measure of slip resistance performance. Our protocol has been approved by the University Health Network Research Ethics Board (UHN, 19-6164, October 2022), with renewals obtained annually, and all participants provided written informed consent prior to participation.

B Dataset

We analyzed data from 18 participants, each of whom tested 4 to 15 commercially available winter footwear models. The selected footwear spans a broad spectrum of outsole designs and material compositions to ensure variability in slip resistance performance. The materials were primarily categorized into three types: 1) rubber; 2) Green Diamond; and 3) Arctic/Icelandic Grip technologies. Arctic/Icelandic Grip technologies are developed for extreme winter performance, incorporating embedded microscopic fibers into polymer blends. These fibers are designed to enhance surface penetration and grip on ice. Specifically, Icelandic Grip integrates silica-infused natural rubber and invisible, vertically embedded glass fibers to improve flexibility and traction in subzero conditions. Due to their similar structural composition and traction mechanisms, Arctic/Icelandic Grip is grouped into a single category for analysis. In contrast, Green Diamond technology introduces embedded crushed industrial diamond particles into rubber compounds. All footwear samples had rubber outsoles as the base material. For example, the footwear shown in Fig. 1(a) consists of only rubber, Fig. 1(b) combines rubber with Green Diamond technology, and Fig. 1(c) integrates Icelandic Grip technology.

The dataset included a total of 578 camera-recorded walking trials that were evenly distributed between dry and wet ice conditions. We also obtained anthropometric data on age, sex,

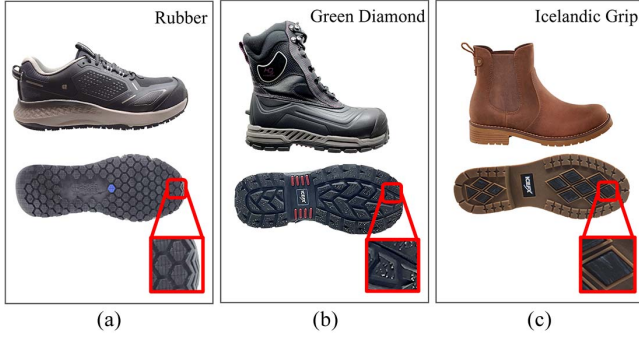


Fig. 1. Outsole materials tested: (a) rubber-only; (b) rubber with Green Diamond; and (c) rubber with both Green Diamond and Icelandic Grip.

height, weight, and BMI of the participants. For our network analysis, age was categorized into two groups: Young Adulthood (approximately 20–35 years) and Middle Adulthood (36–64 years). BMI was also categorized into two groups: 1) normal weight (BMI between 18.5 and 24.9 kg/m²); and 2) overweight (BMI over 25 kg/m²).

C. Preprocessing

Biomechanical features were extracted from video recordings of participants walking on a level surface (0° angle) with their self-selected pace while wearing the test footwear. Skeleton tracking was performed on videos using MediaPipe [41], which overlays 33 full-body keypoints per frame. From these data, we extracted a range of biomechanical features that may influence footwear slip resistance and the MAA testing protocol.

In total, 18 biomechanical parameters were extracted from both the left and right sides of the body. Biomechanical factors include: 1) gait kinematics (e.g., step length and step width); 2) joint angles (e.g., ankle angle, hip angle, and knee angle); 3) center of mass (COM) velocity; and 4) stability-related measures (e.g., lateral sway, vertical shoulder asymmetry, trunk tilt, trunk rotation, and head velocity).

Step length refers to the distance covered between successive heel strikes of opposite feet. Step width, defined as the mediolateral distance between the feet during walking, can capture balance-related gait characteristics. The joint angles, including ankle, knee, and hip at heel strike, were extracted for both the left and right sides when the foot first contacts the ground. Both feet and COM velocity at heel strike were calculated using a central difference method and reflect the participant's forward motion speed at this gait event. Lateral sway describes the side-to-side movement of the body during walking, often linked to balance control. To assess postural control, we extracted additional stability-related features, including vertical shoulder asymmetry as a proxy for trunk stability, along with average trunk tilt, trunk rotation, and head velocity. Vertical displacement of the body's center across frames was also averaged to assess vertical gait dynamics. The extracted features were standardized and resampled to form a consistent sequence of frames per trial.

D. Feature Selection with Network Analysis

To examine the factors contributing to slip risk, we employed a network analysis approach [42] that captures both linear and nonlinear relationships among biomechanical and anthropometric variables. This technique allowed us to visualize and quantify complex interdependencies beyond what traditional statistical models can detect.

To minimize variability introduced by footwear differences during feature selection, we focused our analysis on a single footwear model, characterized by a consistent outsole pattern and material. All 18 participants tested this footwear under both wet and dry ice conditions. All anthropometric variables and biomechanical parameters were included as input in our network analysis. For biomechanical variables, we obtained the average of values over frames for each video.

We built a hybrid network to capture both linear and nonlinear relationships between features. Linear relationships were identified using Pearson correlation, and only strong correlations (above 75%) were included. For nonlinear relationships, we used mutual information (MI) to measure the strength of association between variables. Instead of using a fixed threshold for MI, we applied a data-driven method to determine which connections to include in the network. This approach allowed us to explore how different features interact with one another in predicting MAA. We calculated MI between every pair of variables in our dataset as shown in (1) [43] and then examined the distributions

$$I(X, Y) = \int \int p(x, y) \log \left(\frac{p(x, y)}{p(x)p(y)} \right) dx dy. \quad (1)$$

We set the MI threshold to the median value of all MI values in our dataset. We then combined the correlation-based and MI-based networks to create a unified network that captures both linear and nonlinear relationships. This integrated approach enabled a more comprehensive and data-driven method for feature selection.

To identify the most relevant features, we used PageRank algorithm, which ranks variables based on the strength and influence of their connections within the feature network [44], [45]. Unlike simple correlation-based measures, PageRank considers both the number and quality of a feature's associations, highlighting those that are not only well-connected but also linked to other influential features. This makes it particularly effective for selecting features that play a central role in the overall structure of the data. Based on this analysis, we selected the top similar features with the highest PageRank scores under wet and dry ice conditions.

E. Transformer-Based Prediction System

To evaluate whether biomechanical factors can be useful to predict slip resistance performance and personalize it based on individual gait patterns, we incorporated these features into a transformer-based model. The model captures temporal dynamics in gait and posture, allowing us to account for how individual biomechanical traits influence slip risk. This study extends prior human-centered and AI-based slip safety research

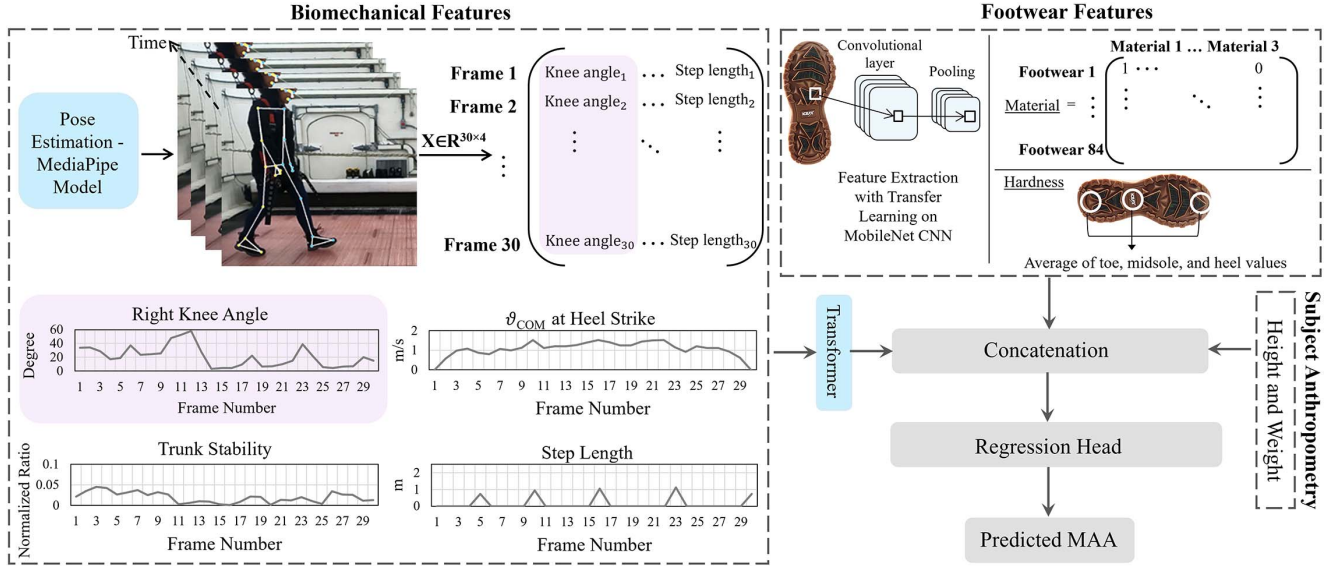


Fig. 2. Flowchart of the transformer-based regression model. Selected temporal biomechanical features from gait data are standardized, resampled to 30 frames, and embedded into a 64-dimensional latent space with added positional encodings.

by introducing transformer-based temporal modeling and multimodal data fusion of biomechanical data and footwear outsole characteristics. The model architecture has been shown in the top part of Fig. 2. For each trial, we extracted:

1) *Temporal Biomechanical Features*: The selected four time series features from network analysis were standardized and resampled to a fixed length of 30 frames (uniformly sampled from the walking sequence). We selected 30 frames because the minimum duration of the videos was 32 frames. Fig. 2 presents a sample of each feature extracted from 18 unique participants wearing 84 different types of footwear, achieving MAA ranging from 1° to 15° .

A linear embedding layer projected each biomechanical frame into a 64-dimensional latent space. Sinusoidal positional encodings were added to retain the temporal order of frames. A two-layer transformer encoder with four attention heads per layer processed the embedded sequence. Temporal mean pooling was applied across frames to produce a fixed-size representation of each trial's gait pattern.

2) *Footwear Outsole Features*: We provided an image of the footwear outsole as a separate input modality to capture visual features, particularly tread patterns. These images were processed using a convolutional neural network (CNN) through transfer learning on a MobileNet architecture [46], pretrained on the ImageNet dataset [47]. MobileNet was chosen as it had previously outperformed other common CNN architectures on a similar task [48]. The CNN output was flattened and passed through a fully connected layer to produce a 64-dimensional feature vector, matching the dimensionality of the biomechanical feature representations from the transformer encoder. We also used outsole hardness and sole material type (encoded categorically) as additional features to our model.

We used a durometer to obtain Shore A hardness, which is the measure of firmness on a 0–100 scale at three locations on each

shoe's tread: 1) heel; 2) midfoot; and 3) toe regions. Measurements were taken on both left and right soles. We then averaged across all three regions and both sides to generate a single representative hardness value for each footwear. This averaging was justified by the low within-shoe variability observed across regions (standard deviation ± 1.7 Shore A units on a 0–100 scale), indicating relatively uniform material properties within each footwear model.

3) *Anthropometry Features*: Participants' height and weight were identified as key features through our network analysis and are incorporated into the model alongside temporal biomechanical and footwear outsole features, as illustrated in Fig. 2.

Our model consisted of a fully connected feedforward neural network with ReLU activation and dropout, designed to predict the continuous MAA values. The model was trained using root mean squared error (RMSE) as a loss function, with 10-fold cross-validation and early stopping applied during hyperparameter tuning. Final performance was assessed using RMSE and R^2 score. To ensure robust generalization, we further conducted an outer 10-fold cross-validation, allowing each data point to serve as an unseen test instance at least once. All biomechanical and footwear features were standardized before training. This modeling has been conducted separately on wet and dry ice conditions.

III. RESULTS

Our analysis was conducted on 18 (nine males and nine females) participants with an average age of 28.9 ± 6.2 years, height of 168.7 ± 8.4 cm, and weight of 72.1 ± 12.5 kg. Based on BMI classification, nine participants had a normal weight, and nine were categorized as overweight. In terms of age groups, 14 participants fell into the Young Adulthood category, while four were classified as Middle Adulthood.

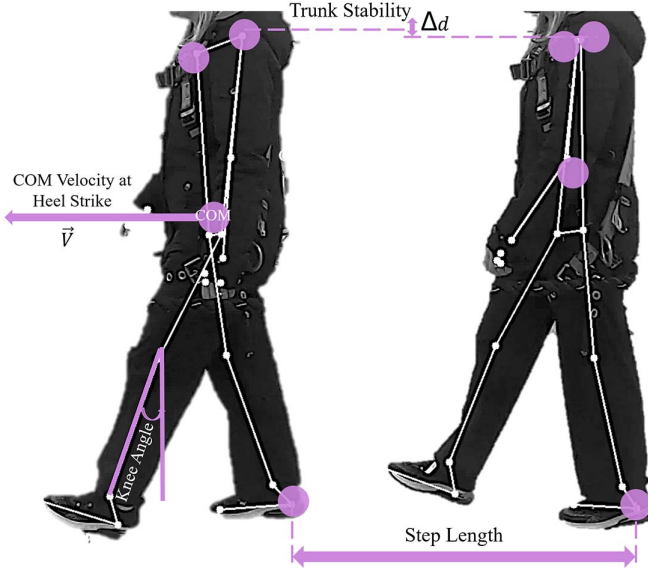


Fig. 3. Key variables of right knee angle, COM velocity, step length, and trunk stability consistently showed strong importance with slip resistance.

The network analysis was conducted on anthropometric (age, BMI, sex, weight, height) and biomechanical variables for wet and dry ice conditions separately to identify relationships and community structures. We found the median of MI as 0.12. The analysis showed community clusters among the variables and identified the most influential factors based on PageRank scores. Across both dry and wet conditions, right knee angle, COM velocity at heel strike, step length, trunk stability, weight, and height consistently emerged as key features associated with slip resistance performance (Fig. 3).

The network analysis revealed three biomechanically coherent clusters: 1) a trunk stability cluster comprising trunk tilt, trunk rotation, and vertical displacement, which covary as part of the body's axial postural control strategy; 2) a lower limb kinematics cluster encompassing hip, knee, and ankle angles, reflecting mechanical coupling along the kinematic chain; and 3) a locomotion dynamics cluster anchored by COM velocity alongside foot velocities and lateral sway, which functions as an integrative output of both upstream subsystems. The interaction between clusters (1) and (3) is particularly relevant to model interpretability: COM velocity is jointly regulated by trunk stability and lower limb propulsion, such that disruptions in trunk control, as seen under challenging surface conditions, reduce the consistency of forward COM progression and increase lateral sway.

Following hyperparameter fine-tuning, the transformer-based model was trained using the Adam optimizer with a learning rate of $1e^{-4}$. To prevent overfitting, regularization techniques such as a dropout rate of 0.3 and weight decay of $1e^{-5}$ were applied. Early stopping was used with a patience of 15 epochs based on validation loss.

As expected, each individual exhibited distinct gait parameters when wearing different types of footwear. An illustrative example is presented in Fig. 4, where a single participant

wore three footwear types with slip resistance scores of 10° , 13° , and 15° , respectively. Due to these inter-individual differences, we employed stratified 10-fold cross-validation as our primary evaluation strategy to ensure representation of diverse footwear–gait combinations across folds, which aligns with the study's objective of personalized slip resistance assessment. In the example shown in Fig. 4, the participant achieved a high MAA score of up to 15° , which corresponded with a lower knee angle, shorter step length, improved trunk stability, and lower velocity of COM at heel strike. These results highlighted the importance of biomechanical factors on MAA scores.

The developed multimodal architecture was evaluated on 289 walking trials collected from the 18 participants while testing 82 unique footwear types under both wet and dry ice conditions (Table I). The model's performance varied notably between the two ice conditions. The model achieved a higher mean RMSE of $2.34^\circ \pm 0.43^\circ$ on wet ice, indicating greater prediction error compared with dry ice, which had a lower mean RMSE of $1.34^\circ \pm 0.34^\circ$. Similarly, the MAE was higher in wet condition ($1.92^\circ \pm 0.43^\circ$) than in dry ($1.05^\circ \pm 0.28^\circ$), suggesting more deviation from actual values on wet surfaces.

Interestingly, the R^2 scores were relatively similar between the two conditions, showing that the model explained a comparable proportion of variance in both scenarios. The MAPE (%) was notably higher in the wet condition ($21.45\% \pm 6.89\%$) compared with dry ($10.48\% \pm 3.58\%$), suggesting that relative prediction errors were more pronounced under wet ice conditions. These results indicate that although the model performs reasonably well under both conditions, its performance is notably reduced on wet ice, likely due to greater variability and lower slip predictability on such surfaces.

We also showed the RMSE metrics as boxplots across the 10 folds, as shown in Fig. 5. The interquartile ranges and whisker lengths in the RMSE plots indicated considerable fold-to-fold variability, particularly under wet condition, suggesting that model performance was less stable across different data subsets. A similar pattern was observed in the MAE measurements, with wet condition showing higher absolute deviations from the predicted values and greater variability across folds. The impact of surface conditions was most evident in the MAPE results, with wet condition producing percentage errors ranging from 15% to 30%, compared with the more acceptable 5% to 15% range observed under dry condition.

IV. DISCUSSION

Our transformer-based model builds upon recent advances in AI-driven slip safety prediction [48], [49], [50]. Sarıışık and Çoşkun successfully applied machine learning (XGBoost, K-Nearest Neighbors, Support Vector Classifier) to predict dynamic CoF in public service buildings based on floor surface materials, environmental conditions, and testing methods. This approach achieved high classification performance (F1 score: 0.75, AUC: 0.83) by identifying environmental conditions as the most influential predictor of slip risk. However, their model operates at the floor-surface level, generating safety classifications (unsafe, conditionally safe, safe, very safe) based on

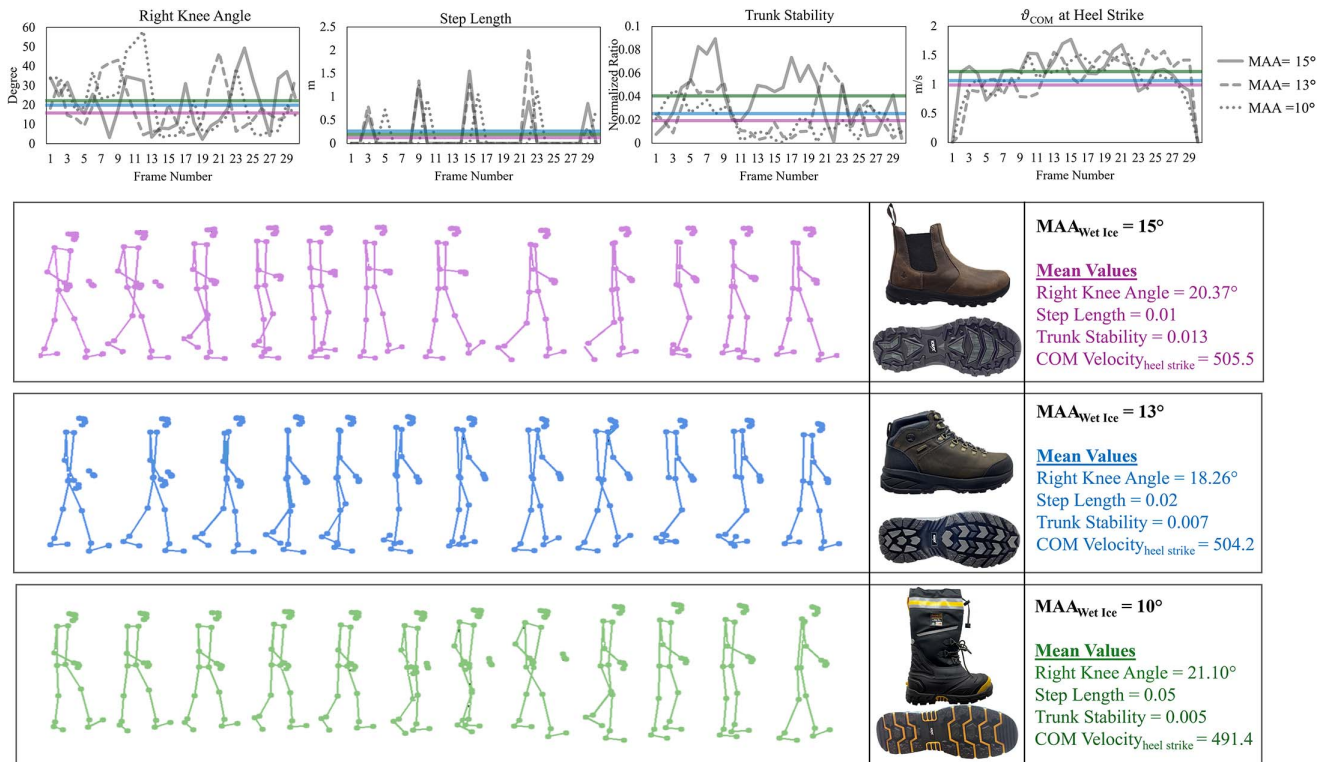


Fig. 4. Twenty-seven-year-old male participant (height: 175 cm, BMI: 25.5 kg/m², categorized as overweight and young) walking on wet ice with three types of footwear.

TABLE I
PERFORMANCE METRICS UNDER WET AND DRY ICE CONDITIONS CONSIDERING 10 TEST FOLDS

Condition	Max RMSE (°)	Min RMSE (°)	Mean RMSE (°)	Max MAE (°)	Min MAE (°)	Mean MAE (°)	MAPE (%)	R ²
Wet Ice	2.91	1.44	2.34 ± 0.43	2.46	1.09	1.92 ± 0.43	21.45 ± 6.89	0.47 ± 0.12
Dry Ice	1.83	0.78	1.34 ± 0.34	1.53	0.62	1.05 ± 0.28	10.48 ± 3.58	0.45 ± 0.09

aggregated dynamic CoF values without incorporating individual biomechanical differences [50]. In contrast, the choice of transformer architecture over traditional machine learning models offers advantages in capturing long-range dependencies to learn how movement stability evolves during walking, which is critical for predicting slip events.

Our transformer-based model operates on biomechanical and outsole image features largely as abstract inputs. However, the learned features fundamentally encode physical tribological mechanisms that govern slip resistance at the footwear-ice interface. The CNN-based outsole image encoder learns visual features that directly correspond to tread pattern geometry. The transformer encoder processes time-series gait data to capture dynamic friction transitions during heel strike and stance phase. These features are aligned well with tribology mechanisms that connect physical friction mechanisms with safety outcomes [8], [14], [15], [18], [19], [23], [26], [27], [51].

This study serves as proof of concept, showing that a vision-based biomechanical system, when combined with footwear properties, can effectively assess slip risk of footwear. Unlike previous approaches that relied on wearables [28] or laboratory-based motion capture systems [52], the proposed vision-based

approach offers a less invasive, more scalable, and potentially more practical solution for deployment in real-world environments. In a previous study on estimating footwear slip resistance based solely on footwear characteristics using a similar dataset, the model achieved RMSE values of 2.31° under dry ice and 3.05° under wet ice conditions [53]. The relatively lower prediction errors achieved by our proposed multimodal model in this article suggest that incorporating biomechanical factors significantly improves the accuracy of slip resistance score estimation. Therefore, ensuring diversity in participants' biomechanical and anthropometric characteristics is crucial for developing robust and generalizable slip resistance prediction models. This helps ensure that the results of human-centered tests e.g., MAA are conservative, generalizable, and reflective of a broader population.

For further comparison, we performed a LOSO cross-validation analysis as a complementary evaluation strategy. In leave-one-subject-out (LOSO) cross-validation, excluding an entire participant during training would hinder the model's ability to generalize across different footwear types, thereby limiting its capacity to learn footwear-specific biomechanical patterns essential for accurate slip resistance prediction. However,

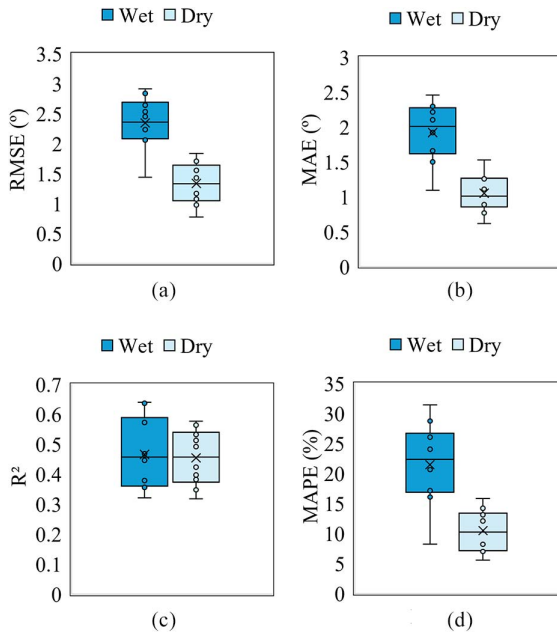


Fig. 5. Boxplot of (a) RMSE, (b) MAE, (c) R^2 , and (d) MAPE(%) in wet and dry conditions across 10-fold cross-validation.

to address potential concerns regarding generalization to unseen participants and learning person-specific patterns, we also conducted LOSO cross-validation as a complementary analysis. In LOSO validation, the model was trained on 17 participants and tested on the remaining held-out participant across all 18 iterations, ensuring complete separation of individual biomechanical patterns between training and test sets. The LOSO validation revealed minimal performance degradation compared with stratified 10-fold cross-validation, with RMSE increasing from 1.34° to 1.42° (+6.2%) under dry ice conditions and from 2.34° to 2.48° (+6.1%) under wet ice conditions. R^2 remained largely stable, decreasing only slightly from 0.45 to 0.42 (−6.4%) under dry conditions and from 0.47 to 0.44 (−5.7%) under wet conditions. These results indicate that the model generalizes effectively to completely unseen individuals, with less than 10% performance loss across all metrics.

Wet ice creates a thin liquid water film at the footwear-ice interface that fundamentally alters the tribological dynamics compared with dry ice. This water film reduces the mechanical interlocking between the outsole tread and the ice surface, introducing greater unpredictability in the frictional forces. As a result, the relationship between these gait parameters and the resulting MAA score becomes less deterministic under wet conditions, which directly contributes to the near-doubling of RMSE ($1.34^\circ \pm 0.34^\circ$ dry versus $2.34^\circ \pm 0.43^\circ$ wet) and MAPE ($10.48\% \pm 3.58\%$ dry versus $21.45\% \pm 6.89\%$ wet).

Bland-Altman analysis shown in Fig. 6 also confirmed superior agreement under dry condition with minimal systematic bias and tighter limits of agreement, whereas wet condition demonstrated wider scatter and potential proportional bias. Under wet condition, as shown in Fig. 6(a), the difference versus mean plot showed considerable scatter, with prediction

differences ranging from approximately -6° to $+6^\circ$. This indicates substantial variability in the model's performance under wet condition. In contrast, Fig. 6(b) demonstrated superior agreement, with data points clustering more tightly around the zero-difference line and exhibiting narrower 95% limits of agreement. The mean difference line in dry condition remained closer to zero, indicating minimal systematic bias in predictions.

Based on internal analyses conducted at KITE-UHN, an absolute error of 1° in the MAA score is considered an acceptable threshold for predictive accuracy. The gray box in Fig. 6 highlights data points with errors between -1° and $+1^\circ$. As illustrated in these figures, 69% (200 out of 289) of predictions for wet condition fell outside of the acceptable range of -1° to $+1^\circ$, compared with 40% (117 out of 289) under dry condition. Among these, the model made conservative predictions (errors below -1°) in 35% of the large errors (70 cases out of 200) under wet condition and 33% (39 cases out of 117) under dry condition, as highlighted in blue in Fig. 6.

The scatter plots in Fig. 7 illustrate the model's performance characteristics across the full range of actual MAA values under both ice conditions. Under wet ice conditions, the model demonstrates that systematic overestimation is particularly pronounced in the mid-range MAA values (8° – 13°). In contrast, the dry ice condition reveals markedly different performance characteristics. The model shows better tracking along the line of perfect agreement, particularly for lower MAA values (6° – 10°). However, some underprediction becomes evident at higher actual MAA values (above 12°).

So far, most modeling research using AI or finite element methods (FEM) to predict slip resistance scores has focused solely on footwear properties [48], [49], [51], [54], [55]. This study demonstrates that incorporating key biomechanical variables temporally, beyond just the footwear characteristics, can significantly improve prediction performance. This approach highlighted the importance of developing personalized slip prevention strategies for footwear testing based on individual movement patterns, rather than relying solely on the mechanical properties of footwear. In addition, since the MAA score is defined as the minimum value among the MAAs measured from four participants, it is crucial to include individuals with diverse biomechanical profiles to obtain a more accurate and representative MAA score.

V. LIMITATIONS AND FUTURE WORK

This study has several limitations. First, the participant pool was limited in age range, with most individuals being under 30 years old. This may restrict the generalizability of the findings, especially for older adults who are at greater risk of slips and falls. Additionally, factors such as prior awareness of a potential slip and previous slip experiences, which can influence gait, were not controlled for. Fatigue is another important consideration, as it can lower gait stability and increase slip risk, but it was not specifically assessed or induced in this study. Future work should aim to include a more diverse age range, assess the role of fatigue, and account for psychological and

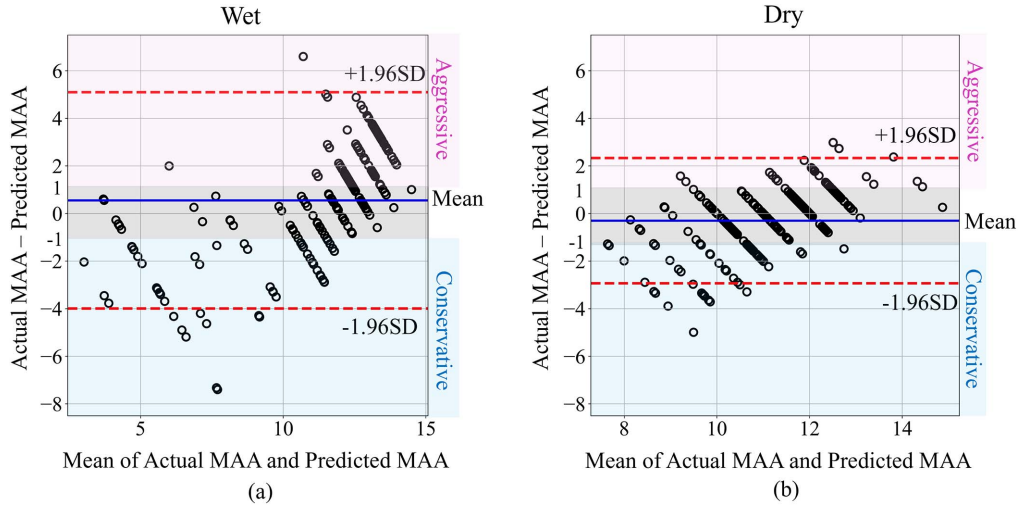


Fig. 6. Bland-Altman plots showing agreement between actual and predicted MAA values under: (a) wet; and (b) dry conditions. The solid blue line represents the mean difference, while the dashed red lines indicate the limits of agreement (± 1.96 standard deviations). The gray box represents data points with errors between -1 and $+1$. The blue region highlights the proportion of conservative predictions (predicted values lower than actual), while the purple region shows aggressive predictions (predicted values higher than actual).

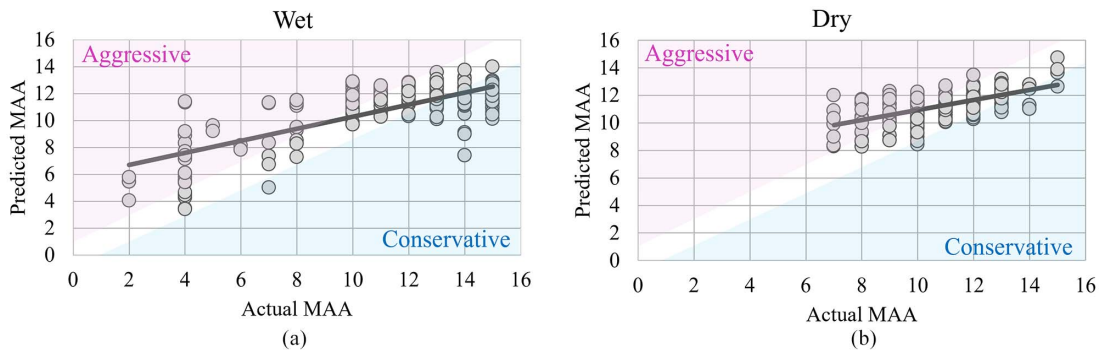


Fig. 7. Scatter plots showing model performance for: (a) wet; and (b) dry ice conditions. The white area represents data points with errors between -1 and $+1$. The blue region highlights the proportion of conservative predictions (predicted values lower than actual), while the purple region shows aggressive predictions (predicted values higher than actual).

experiential factors to improve the robustness and real-world applicability of the results. Our fixed 30-frame resampling approach assumes proportional representation of gait events across participants. While our event-based feature extraction and model validation suggest this assumption holds reasonably well, future work could investigate time-warping or variable-length sequence methods to better preserve the absolute timing of critical gait events such as heel strike.

VI. CONCLUSION

We developed a multimodal deep learning system that combined time-series gait data, footwear material features, and footwear images for MAA prediction. The model demonstrated condition-dependent predictive performance, achieving superior accuracy under dry condition (RMSE: 1.2° – 1.8° , R^2 : 0.4 – 0.6 , MAPE: 5% – 15%) compared with wet condition (RMSE: 2.2° – 2.8° , R^2 : 0.4 – 0.6 , MAPE: 15% – 30%). While the model maintained consistent explanatory power across both environments, the substantial increase in prediction errors and

cross-validation variability under wet condition revealed the complex challenges of slip risk assessment in moisture-prone environments. These findings highlighted the critical need for adaptive and context-aware testing protocols that account for environmental conditions and opened new avenues for intelligent footwear assessment tools that incorporate condition-specific calibration for more reliable injury prevention applications.

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